

Workbook Standard Functions

Standard functions include standard arithmetic and conditional functions, such as SUM() and IF (), and in most cases are also found in spreadsheets such as Microsoft Excel. Standard functions entered in workbook cells can be modified by selecting the cell and editing the text in the Cell Information Panel or by double clicking the cell and editing directly.

The standard functions supported in workbooks can be divided into groups as follows:

- [Arithmetic functions](#)
- [Statistical functions](#)
- [Logical functions](#)
- [Lookup functions](#)
- [Date and Time functions](#)
- [Text functions](#)
- [Financial functions](#)
- [IS functions](#)

* - Indicates the function is specific to CostX and generally not supported by other spreadsheet programs such as Microsoft Excel.

Arithmetic functions

A set of mathematical and trigonometric functions.

Standard Formula	Examples	Description
ABS	=ABS(B6)	Returns the absolute value.
ACOS	=ACOS(B16)	Returns the arccosine in radians.
ACOSH	=ACOSH(A1)	Returns the inverse hyperbolic cosine in radians.
ASIN	=ASIN(B6)	Returns the arcsine in radians.
ASINH	=ASINH(C2)	Returns the inverse hyperbolic sine of the argument in radians.
ATAN	=ATAN(C5)	Returns the arctangent in radians.
ATAN2	=ATAN2(A1, A2)	Returns the arctangent using x- and y-coordinates in radians.
ATANH	=ATANH(D4)	Returns the inverse hyperbolic tangent in radians.
CEILING	=CEILING(2.5, 1)	Rounds the first argument up to the nearest multiple based on the significance specified by the second argument.
COS	=COS(0.5)	Returns the cosine. Takes one argument in radians.
COSH	=COSH(D7)	Returns the hyperbolic cosine. Takes one argument in radians.

Standard Formula	Examples	Description
COUNTIF	=COUNTIF(A1:A7,">2")	Two arguments: the range of cells and the condition of type string. Counts the number of non blank cells within a range which meets the given condition.
DACOS*	=DACOS(B16)	Returns the arccosine in degrees.
DACOSH*	=DACOSH(A1)	Returns the inverse hyperbolic cosine in degrees.
DASIN*	=DASIN(B6)	Returns the arcsine in degrees.
DASINH*	=DASINH(C2)	Returns the inverse hyperbolic sine of the argument in degrees.
DATAN*	=DATANH(C5)	Returns the arctangent in degrees.
DATAN2*	=DATAN2(A1,A2)	Returns the arctangent using x- and y-coordinates in degrees.
DATANH*	=DATANH(D4)	Returns the inverse hyperbolic tangent in degrees.
DCOS*	=DCOS(60)	Returns the cosine. Takes one argument in degrees.
DCOSH*	=DCOSH(30)	Returns the hyperbolic cosine. Takes one argument in degrees.
DSIN*	=DSIN(30)	Returns the sine. Takes one argument in degrees.
DSINH*	=DSINH(C2)	Returns the hyperbolic sine. Takes one argument in degrees.
DTAN*	=DTAN(45)	Returns the tangent. Takes one argument in degrees.
DTANH*	=DTANH(30)	Returns the hyperbolic tangent. Takes one argument in degrees.
DEGREES	=DEGREES(PI()/2)	One argument that specifies angle in radians. Converts radians to degrees.
EVEN	=EVEN(-1)	Rounds the argument up to the nearest even integer.
EXP	=EXP(C1)	Returns the exponent value of the argument.
FACT	=FACT(2)	Returns the factorial of the argument. The argument must be nonnegative.
FLOOR	=FLOOR(-2.5, -2)	Rounds the first argument down, towards zero, to the nearest multiple of the significance specified by the second argument.
INT	=INT(E4)	Rounds the argument down to the nearest integer.
LN	=LN(C1)	Returns the natural logarithm.

Standard Formula	Examples	Description
LOG	=LOG(100, 10)	Two arguments: the number and the base. Returns the logarithm of a number to the specified base.
LOG10	=LOG10(1000)	Returns the base-10 logarithm of the argument.
MOD	=MOD(4,3)	Two arguments: a number and divisor. Returns the remainder after the number is divided by the divisor.
ODD	=ODD(3.5)	Rounds the argument up to the nearest odd integer.
PI	=PI()	No arguments required. Returns the value of Pi.
POWER	=POWER(A1,4)	Two arguments: the number and the power. Raises the number to the base.
PRODUCT	=PRODUCT(A1:B1) =PRODUCT(A1,C1,3)	Multiplies the comma separated list of arguments ignoring text and blank cells. Arguments can be a cell, number, or range.
RADIANS	=RADIANS(180)	One argument that specifies the angle in degrees. Converts degrees to radians.
RAND	=RAND()	No arguments required. Returns a random number between 0 and 1.
RONE	=RONE(2.44)	* Takes one argument. If the argument is 0, 0 is returned, otherwise it rounds up the argument to the nearest integer and adds 1.
ROUND	=ROUND(20.57, 1)	Two arguments: the number and the number of digits. Rounds the first argument to the specified number of digits.
ROUNDDOWN	=ROUNDDOWN(2.75,0)	Rounds the argument toward zero. The first argument specifies a number to round down. The second argument defines the number of digits to which you wish to round the first argument.
ROUNDUP	=ROUNDUP(-3.42,1)	Rounds the argument toward infinity. The first argument specifies a number to round up. The second argument defines the number of digits to which you wish to round the first argument.
SIGN	=SIGN(A2)	Returns the sign.
SIN	=SIN(A1)	Returns the sine. Takes one argument in radians.
SINH	=SINH(1)	Returns the hyperbolic sine. Takes one argument in radians.
SQRT	=SQRT(B5)	Returns the square root.
SUBTOTAL	=SUBTOTAL(H1:H12)	* Takes a list of arguments, (typically a range of cells). Sums all the values in the list. More Details...

Standard Formula	Examples	Description
SUM	=SUM(A1:C12) =SUM(A1, 3.14, 1.57)	Takes a list of arguments. Sums all the values in the list.
SUMIF	=SUMIF(A1:A10, ">50", C1:C10)	Sums the values in a range of cells based on a single criterion (for example, is larger than 50). Takes three arguments: 1) range : the range of cells that you wish to apply the criterion against. 2) criterion_range : the criterion used to determine which cells to add. 3) sum_range : (Optional) The range of cells to sum together. If this parameter is omitted, range will be used as the sum_range .
SUMSQ	=SUMSQ(B1:D1) =SUMSQ(1, 2, 3, 4)	Takes a list of arguments. Sums the square of values in the list.
TAN	=TAN(C3)	Returns the tangent. Takes one argument in radians.
TANH	=TANH(0)	Returns the hyperbolic tangent. Takes one argument in radians.
TRUNC	=TRUNC(PI())	The first argument is the number you wish to truncate. The second is optional and is the number of decimal places to truncate at (0 is default). Returns the number truncated at the decimal place specified. Compare this with the Int function which rounds down to the smallest integer rather than just cutting off digits.

Statistical functions

Basic statistical functions.

Standard Formula	Examples	Description
AVERAGE	=AVERAGE(A1:A5) =AVERAGE(10.3, 9.1)	Takes a list of arguments. Calculates the average value of values within the list.
AVERAGEA	=AVERAGEA(A1:A5)	Takes a list of arguments. Calculates the average value of the non-empty cells referenced. AVERAGEA(range) equals SUM(range)/COUNTA(range).
COUNT	=COUNT(A1:A4) =COUNT(1, 5, 8)	Takes a list of arguments. Returns the number of cells in a given range.
COUNTA	=COUNTA(A1:A4)	Takes a list of arguments. Counts the number of non-empty cells in a given range.

Standard Formula	Examples	Description
COUNTBLANK	=COUNTBLANK(A1:E1)	Takes a list of arguments. Counts the number of empty cells.
MAX	=MAX(A1:D1) =MAX(A1, 100, C2)	Takes a list of arguments. Returns the largest argument value.
MIN	=MIN(A1:D1) =MIN(0, C2)	Takes a list of arguments. Returns the smallest argument value.
SUM	=SUM(A1:C12) =SUM(A1, 3.14, 1.57)	Takes a list of arguments. Sums all the values in the list.
SUMSQ	=SUMSQ(B1:D1) =SUM(1, 2, 3, 4)	Takes a list of arguments. Sums the square of values in the list.

Logical functions

Logical functions take logical values as arguments and return a logical value as a result.

Standard Formula	Examples	Description
AND	=AND(1<B4, B4<100)	Takes a list of arguments. Logical AND operation. Returns TRUE if all values within the list are TRUE; returns FALSE if one or more values within the list evaluates to FALSE.
FALSE	=FALSE()	No arguments required. Returns the logical value FALSE.
IF	=IF(A1>100,1,0) =IF(A10<=100, "Within budget", "Over budget")	Takes three arguments: the logical expression, the value to return if the expression succeeds, the value to return the expression fails. Returns the second argument if the logical expression evaluates to TRUE and the third argument otherwise.
NOT	=NOT(1>D4)	One argument of logical type. Logical NOT operation. Reverses the value of its argument.
OR	=OR(A1>=10, A1<=10)	Takes a list of arguments. Logical OR operation. Returns TRUE if any argument is TRUE; returns FALSE if all values within the list evaluate to FALSE.
TRUE	=TRUE()	No arguments required. Returns the logical value TRUE.

Lookup functions

Standard Formula	Examples	Description
VLOOKUP	=VLOOKUP(2,I1:J6,2)	

Standard Formula	Examples	Description
	=VLOOKUP ("C",I1:M6,5,FALSE) =VLOOKUP(A12,Calc! A1:B6,2)	Takes three arguments: the value to search for, the range or array of cells to search in, and the column in the array from which to return the result. Searches for the stipulated value in the first column of the array, and returns the corresponding value from the specified column on the same row of the array. Optionally a fourth argument can be added to specify the required match as TRUE or FALSE. TRUE (or omitted) specifies that an approximate match is acceptable. If an exact match is not found, the largest value that is less than search value is used. FALSE specifies that only an exact match is acceptable. If there are two or more values in the first column of the array that match the search value, the first value found is used. More Details...
HLOOKUP	=HLOOKUP(2,I1:P5,2) =HLOOKUP("C",I1:P5,5) =HLOOKUP(A12,Calc! A1:B6,2)	Takes three arguments: the value to search for, the range or array of cells to search in, and the row in the array from which to return the result. Searches for the stipulated value in the first row of the array, and returns the corresponding value from the specified row on the same column of the array. Optionally a fourth argument can be added to specify the required match as TRUE or FALSE. TRUE (or omitted) specifies that an approximate match is acceptable. If an exact match is not found, the largest value that is less than search value is used. FALSE specifies that only an exact match is acceptable. If there are two or more values in the first row of the array that match the search value, the first value found is used. More Details...

Date and Time functions

Standard Formula	Examples	Description
DATE	=DATE(1900,1,1)	Three arguments defining the year, month and day. Calculates the serial number that represents a specified date.
DAY	=DAY(TODAY())	One argument defining the serial number of the required date. Returns the day portion of a given date.
HOUR	=HOUR(NOW())	One argument defining the serial number of the required date/time value. Returns the hour portion of a given date/time value.

Standard Formula	Examples	Description
MONTH	=MONTH(TODAY())	One argument defining the serial number of the required date. Returns the month portion of a given date.
MINUTE	=MINUTE(NOW())	One argument defining the serial number of the required date/time value. Returns the minutes portion of a given date/time value.
NOW	=NOW()	No arguments required. Returns the current time in general format. You can apply further formatting to the result of the function.
SECOND	=SECOND(NOW())	One argument defining the serial number of the required date/time value. Returns the seconds portion of a given date/time value.
TIME	=TIME(16, 48, 10)	Three arguments defining hour, minute and second parts of a time value. Returns a decimal number for a specified time.
TODAY	=TODAY()	No arguments required. Returns the serial number of the current date.
WEEKDAY	=WEEKDAY(TODAY()) =WEEKDAY(DATE(2002, 12, 1), 1) Returns 1(Sunday)	Two arguments: the serial number of the required date, weekday base. Returns the day of the week corresponding to the specified date. The weekday base identifies the first day of the week and determines the return value type: 1 or omitted: the first day of the week is Sunday. The function returns 1 for Sunday, 2 for Monday, etc. 2: the first day of the week is Monday. The function returns 1 for Monday, 2 for Tuesday, etc. 3: the first day of the week is Monday. The function returns 0 for Monday, 1 for Tuesday, etc.
YEAR	=YEAR(TODAY())	One argument defining the serial number of the required date. Returns the year portion of a given date.

Text functions

Standard Formula	Example	Description
& (ampersand)	= "Yellow "&"river"	Two arguments of type string. Concatenates specified strings.
CONCATENATE	=CONCATENATE("CostX", "Workbook")	Takes a list of strings. Joins several text strings in one text string. An alternative to "&".
DOLLAR	=DOLLAR(957.344, 2)	Two arguments defining the value and the number of digits to the right of the decimal point in the output string. Converts the number to text using currency format \$#,##0.00_);

Standard Formula	Example	Description
		(\$#,##0.00), with the decimals rounded to the specified number of places.
FIXED	=FIXED(1234.567, 1)	The first argument is required. The second argument of type integer is optional (the default value is 2). The third argument of type Boolean is optional. Rounds the first argument to the number of decimals determined by the second argument and returns it as a string. The third parameter specifies whether to omit commas in the output string.
LEFT	=LEFT(A1) =LEFT(A1, 3)	The first argument of type string is required. The second argument of type integer is optional. Returns the first character or characters in a text string. The second parameter defines the number of characters to extract. The default value is 1.
LEN	=LEN("ABC")	Takes one argument of type string. Returns the length of a given string.
LOWER	=LOWER(A1)	Takes one argument of type string. Converts a string to lowercase.
MID	=MID("ABC",3,1)	The first argument is of type string, the second and the third arguments are of type integer. Returns the substring of a given text string. The position of the substring is defined by the second parameter. The third parameter specifies the number of characters to extract.
RIGHT	=RIGHT(B2, 3)	The first argument of type string is required. The second argument of type integer is optional. Returns the last character or characters in a text string. The second parameter defines the number of characters to extract. The default value is 1.
TRIM	=TRIM(" CostX Sheet ")	Takes one argument of string type. Removes all spaces from text except for single spaces between words.
UPPER	=UPPER(A1&A2)	Takes one argument of type string. Converts a string to uppercase.

Financial functions

In the financial functions described below, the following parameters are used.

- **Rate** is the interest rate per period. For example, if you have a loan at a 10 percent annual interest rate and make monthly payments, your interest rate per month is 10%/12, or 0.83%. You would enter 10%/12, or 0.83%, or 0.0083, into the formula as the rate.
- **Period** is the period for which you wish to find the interest and must be in the range 1 to NoPeriods.

- **NoPeriods** is the total number of payment periods in an annuity. For example, if you get a four-year car loan and make monthly payments, your loan has 4*12 (or 48) periods. You would enter 48 into the formula for NoPeriods.
- **Payment** is the payment made each period and cannot change over the life of the annuity. Typically, Payment includes principal and interest but no other fees or taxes. For example, the monthly payments on a \$10,000, four-year car loan at 12 percent are \$263.33. You would enter -263.33 into the formula as the Payment.
- **FV** is the future value, or a cash balance you wish to attain after the last payment is made. If FV is omitted, it is assumed to be 0 (the future value of a loan, for example, is 0). For example, if you wish to save \$50,000 to pay for a special project in 18 years, then \$50,000 is the future value. You could then make a conservative guess at an interest rate and determine how much you must save each month.
- **Type** is the number 0 or 1 and indicates when payments are due. Use 0 if the payment is due at the end of the period. Use 1 if payments are due at the start of the period.

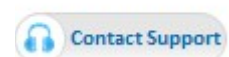
Standard Formula	Usage and Example	Description
PV	=PV (Rate,NoPeriods,Payment,FV,type) =PV(0.83%,120,-1000,0,0) =PV(5%/12,240,500,130000,1)	Required parameters are shown in Bold. Returns the present value of an investment. The present value is the total amount that a series of future payments is worth now. For example, when you borrow money, the loan amount is the present value to the lender.
FV	=FV (Rate,NoPeriods,Payment,PV,type) =FV(0.83%,120,-1000,0,0) =FV(5%/12,240,500,130000,1)	Required parameters are shown in Bold. Returns the future value of an investment based on periodic, constant payments and a constant interest rate.
IPMT	=IPMT (Rate,Period,NoPeriods,PV,FV,type)	Required parameters are shown in Bold. Returns the interest payment for a given period for an investment based on periodic, constant payments and a constant interest rate.
PMT	=PMT(Rate,NoPeriods,PV,FV,type)	Required parameters are shown in Bold. Calculates the payment for a loan based on constant payments and a constant interest rate.
PPMT	=PPMT (Rate,Period,NoPeriods,PV,FV,type)	Required parameters are shown in Bold. Returns the payment on the principal for a given period for an investment based on periodic, constant payments and a constant interest rate.
NPER	=NPER(Rate,Payment,PV,FV,type)	Required parameters are shown in Bold. Returns the number of periods for an investment based on periodic, constant payments and a constant interest rate.

IS functions

Functions which test the type of a value and return a Boolean result.

Standard Formula	Example	Description
ISBLANK	=ISBLANK(A1)	Takes a value of any type. Returns TRUE if cell is empty.
ISERR	=ISERR(H1)	Takes a value of any type. Returns TRUE if the cell contains any error value except #N/A.
ISERROR	=ISERROR(A7)	Takes a value of any type. Returns TRUE if the cell contains any error value (#N/A, #VALUE!, #REF!, #DIV/0!, #NUM!, #NAME?, or #NULL!).
ISLOGICAL	=ISLOGICAL(C3)	Takes a value of any type. Returns TRUE if the specified value refers to a logical value.
ISNA	=ISNA(E5)	Takes a value of any type. Returns TRUE if the cell contains #N/A (value not available) error value.
ISNONTEXT	=ISNONTEXT (D4)	Takes a value of any type. Returns TRUE if the cell does not contain text. Returns TRUE for blank cells.
ISNUMBER	=ISNUMBER(A2)	Takes a value of any type. Returns TRUE if the cell contains a number.
ISTEXT	=ISTEXT(D1)	Takes a value of any type. Returns TRUE if the specified cell contains text.

See also [Workbook Custom Functions](#) and other [Cell Equations](#) topics...



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